PHYDSC402T — Classical Mechanics

Unit 3: Hamiltonian Formulation

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC402T, Full Marks 100 [ESE 70 + CCA 30] | Unit 3: 12 lectures

Notes 2 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

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Top 2 high-mark topics in this unit

1. Hamilton's equations of motion and their derivation from the variational principle — the core derivation of the unit, set as a long question in other universities' papers.
2. Canonical transformations and Poisson brackets, including invariance of Poisson brackets under canonical transformation and equations of motion in Poisson bracket form — two linked derivations repeated across other universities' papers.

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1. The Hamiltonian

In the Lagrangian formulation of mechanics, a dynamical system with n degrees of freedom is described by n generalized coordinates $q_1, q_2, \ldots, q_n$ and their corresponding generalized velocities $\dot{q}_1, \dot{q}_2, \ldots, \dot{q}_f$ in configuration space. To describe the state of the system in terms of coordinates and momenta rather than coordinates and velocities, we define the generalized momentum p_i conjugate to the generalized coordinate q_i as the partial derivative of the Lagrangian $L(q_i, \dot{q}_i, t)$ with respect to $\dot{q}_i$:

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

To transition from the velocity space $(q_i, \dot{q}_i, t)$ to the phase space (q_i, p_i, t), a Legendre transformation is performed. This transformation defines a new function called the Hamiltonian $H(q_i, p_i, t)$, constructed by taking the sum of the products of each generalized momentum and its corresponding generalized velocity, and subtracting the Lagrangian:

$$H(q_1, \ldots, q_n, p_1, \ldots, p_n, t) = \sum_{j=1}^{n} p_j \dot{q}_j - L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_f, t)$$

Under specific standard physical conditions, the Hamiltonian equals the total mechanical energy of the system. Consider a system where the transformation equations between Cartesian coordinates and generalized coordinates do not contain explicit time dependence. In this case, the kinetic energy T is a homogeneous quadratic function of the generalized velocities:

$$T = \frac{1}{2} \sum_{j=1}^{n} \sum_{k=1}^{n} a_{jk}(q_1, \ldots, q_n) \dot{q}_j \dot{q}_k$$

Euler's theorem for a homogeneous function of degree n in variables x_i states that $\sum_i x_i \frac{\partial f}{\partial x_i} = nf$. Applying Euler's theorem to the kinetic energy T, which is homogeneous of degree 2 in the generalized velocities $\dot{q}_j$, yields:

$$\sum_{j=1}^{n} \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = 2T$$

Assuming the potential energy V depends only on the generalized coordinates q_j and has no dependence on the generalized velocities $\dot{q}_j$, the Lagrangian is expressed as:

$$L = T - V$$

Differentiating the Lagrangian with respect to $\dot{q}_j$, while keeping in mind that V does not depend on $\dot{q}_j$, gives:

$$\frac{\partial L}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j}$$

Substituting the definition of generalized momentum $p_j = \frac{\partial L}{\partial \dot{q}_j}$ into this expression shows that:

$$p_j = \frac{\partial T}{\partial \dot{q}_j}$$

Multiplying both sides by $\dot{q}_j$ and summing over all degrees of freedom $j = 1, \ldots, n$ leads to:

$$\sum_{j=1}^{n} p_j \dot{q}_j = \sum_{j=1}^{n} \dot{q}_j \frac{\partial T}{\partial \dot{q}_j}$$

Applying Euler's result $\sum_{j=1}^{n} \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = 2T$ simplifies the sum to:

$$\sum_{j=1}^{n} p_j \dot{q}_j = 2T$$

Substituting this relation into the definition of the Hamiltonian $H = \sum_{j=1}^{n} p_j \dot{q}_j - L$ gives:

$$H = 2T - L$$

Replacing the Lagrangian L with $T - V$ in this equation results in:

$$H = 2T - (T - V)$$

Simplifying the right-hand side yields:

$$H = T + V$$

Hence, when the potential energy is velocity-independent and the coordinate transformations do not explicitly depend on time, the Hamiltonian H represents the total mechanical energy of the system.

2. Hamilton's Equations of Motion

Hamilton's equations of motion are derived by treating the Hamiltonian $H(q_1, \ldots, q_n, p_1, \ldots, p_n, t)$ as a function of the independent phase space variables q_i, p_i, and explicit time t. Writing the formal total differential of $H(q_i, p_i, t)$ using partial derivatives gives:

$$dH = \sum_{i=1}^{n} \frac{\partial H}{\partial q_i} dq_i + \sum_{i=1}^{n} \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial t} dt$$

Alternatively, the differential dH can be evaluated directly from the defining relation of the Hamiltonian $H = \sum_{i=1}^{n} p_i \dot{q}_i - L$:

$$dH = d\left(\sum_{i=1}^{n} p_i \dot{q}_i \right) - dL$$

Applying the product rule to the differential of the summation term gives:

$$d\left(\sum_{i=1}^{n} p_i \dot{q}_i \right) = \sum_{i=1}^{n} p_i d\dot{q}_i + \sum_{i=1}^{n} \dot{q}_i dp_i$$

The total differential of the Lagrangian $L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_f, t)$ expressed in terms of its arguments is:

$$dL = \sum_{i=1}^{n} \frac{\partial L}{\partial q_i} dq_i + \sum_{i=1}^{n} \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial L}{\partial t} dt$$

Substituting the definition of the generalized momentum $p_i = \frac{\partial L}{\partial \dot{q}_i}$ into the expression for dL gives:

$$dL = \sum_{i=1}^{n} \frac{\partial L}{\partial q_i} dq_i + \sum_{i=1}^{n} p_i d\dot{q}_i + \frac{\partial L}{\partial t} dt$$

Substituting the product rule expansion of $d\left(\sum_{i=1}^{n} p_i \dot{q}_i\right)$ and the expanded differential dL back into the differential expression for dH yields:

$$dH = \sum_{i=1}^{n} p_i d\dot{q}_i + \sum_{i=1}^{n} \dot{q}_i dp_i - \left(\sum_{i=1}^{n} \frac{\partial L}{\partial q_i} dq_i + \sum_{i=1}^{n} p_i d\dot{q}_i + \frac{\partial L}{\partial t} dt \right)$$

Subtracting the common term $\sum_{i=1}^{n} p_i d\dot{q}_i$ simplifies the expression to:

$$dH = \sum_{i=1}^{n} \dot{q}_i dp_i - \sum_{i=1}^{n} \frac{\partial L}{\partial q_i} dq_i - \frac{\partial L}{\partial t} dt$$

Lagrange's equations of motion for a system subject to conservative forces are given by:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

Using $p_i = \frac{\partial L}{\partial \dot{q}_i}$, the time derivative is $\dot{p}_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)$, which transforms Lagrange's equation into:

$$\dot{p}_i = \frac{\partial L}{\partial q_i}$$

Substituting $\frac{\partial L}{\partial q_i} = \dot{p}_i$ into the differential dH gives:

$$dH = \sum_{i=1}^{n} \dot{q}_i dp_i - \sum_{i=1}^{n} \dot{p}_i dq_i - \frac{\partial L}{\partial t} dt$$

Rearranging the terms in order of dq_i, dp_i, and dt yields:

$$dH = -\sum_{i=1}^{n} \dot{p}_i dq_i + \sum_{i=1}^{n} \dot{q}_i dp_i - \frac{\partial L}{\partial t} dt$$

Since the differentials dq_i, dp_i, and dt are independent, the coefficients of corresponding differentials in the two independent expressions for dH must be identical.

Matching the coefficient of dq_i from both expressions gives:

$$\frac{\partial H}{\partial q_i} = -\dot{p}_i$$

Rearranging this relation yields the first set of equations:

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Matching the coefficient of dp_i from both expressions gives:

$$\frac{\partial H}{\partial p_i} = \dot{q}_i$$

Rearranging this relation yields the second set of equations:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

Matching the coefficient of explicit time differential dt gives:

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

The set of equations:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

for $i = 1, 2, \ldots, n$ are known as **Hamilton's equations of motion** (or canonical equations of motion). They constitute a set of $2n$ first-order differential equations in place of the n second-order differential equations of the Lagrangian formulation.

3. Hamilton's Equations from the Variational Principle

Hamilton's equations of motion can be derived from a modified form of Hamilton's principle formulated directly in phase space. In the standard Lagrangian formulation, Hamilton's principle considers variations of the path in configuration space where only the generalized coordinates $q_i(t)$ are varied. In contrast, the modified Hamilton's principle treats the generalized coordinates $q_i(t)$ and their conjugate momenta $p_i(t)$ as independent variables in a $2n$-dimensional phase space.

The action integral I expressed in terms of phase space variables is given by

$$I = \int_{t_1}^{t_2} \left(\sum_{i=1}^{n} p_i \dot{q}_i - H(q_1, \ldots, q_n, p_1, \ldots, p_n, t) \right) dt$$

The modified Hamilton's principle states that the actual trajectory of the system in phase space makes the action integral I stationary with respect to arbitrary independent variations of $q_i(t)$ and $p_i(t)$. Thus, the variation of the action must vanish:

$$\delta I = \delta \int_{t_1}^{t_2} \left(\sum_{i=1}^{n} p_i \dot{q}_i - H(q, p, t) \right) dt = 0$$

To evaluate this variation, we apply the variation operator δ inside the integrand while holding the initial time t_1 and final time t_2 fixed. Taking the variation of the terms inside the integral gives

$$\delta I = \int_{t_1}^{t_2} \sum_{i=1}^{n} \left(\delta p_i \dot{q}_i + p_i \delta \dot{q}_i - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right) dt$$

The variation of the velocity coordinate $\delta \dot{q}_i$ is equal to the time derivative of the variation of the position coordinate, so we can write

$$\delta \dot{q}_i = \frac{d}{dt}(\delta q_i)$$

Substituting this relation into the second term of the integrand yields

$$\delta I = \int_{t_1}^{t_2} \sum_{i=1}^{n} \left(\delta p_i \dot{q}_i + p_i \frac{d}{dt}(\delta q_i) - \frac{\partial H}{\partial q_i}\delta q_i - \frac{\partial H}{\partial p_i}\delta p_i \right) dt$$

We integrate the term containing $\frac{d}{dt}(\delta q_i)$ by parts:

$$\int_{t_1}^{t_2} p_i \frac{d}{dt}(\delta q_i) dt = \left[p_i \delta q_i \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \dot{p}_i \delta q_i dt$$

Under the modified Hamilton's principle, the variations of the generalized coordinates are fixed at the endpoints t_1 and t_2, which means

$$\delta q_i(t_1) = 0 \quad \text{and} \quad \delta q_i(t_2) = 0$$

However, the variations of the conjugate momenta $\delta p_i(t)$ are completely free at the endpoints as well as in the interior, because no time derivatives of p_i appear in the action integral integrand. Substituting the endpoint condition $\delta q_i(t_1) = \delta q_i(t_2) = 0$ into the boundary term results in

$$\left[p_i \delta q_i \right]_{t_1}^{t_2} = 0$$

Substituting the integrated-by-parts expression back into the variation of the action integral gives

$$\delta I = \int_{t_1}^{t_2} \sum_{i=1}^{n} \left(\delta p_i \dot{q}_i - \dot{p}_i \delta q_i - \frac{\partial H}{\partial q_i}\delta q_i - \frac{\partial H}{\partial p_i}\delta p_i \right) dt$$

Grouping the terms proportional to the independent variations δq_i and δp_i leads to

$$\delta I = \int_{t_1}^{t_2} \sum_{i=1}^{n} \left[\left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right) \delta p_i - \left(\dot{p}_i + \frac{\partial H}{\partial q_i} \right) \delta q_i \right] dt = 0$$

Since the variations $\delta q_i(t)$ and $\delta p_i(t)$ are arbitrary and mutually independent for all intermediate times $t \in (t_1, t_2)$, the coefficients of δp_i and δq_i inside the integral must independently vanish for every $i = 1, 2, \ldots, n$. Setting the coefficient of δp_i to zero gives

$$\dot{q}_i - \frac{\partial H}{\partial p_i} = 0 \implies \dot{q}_i = \frac{\partial H}{\partial p_i}$$

Setting the coefficient of δq_i to zero gives

$$\dot{p}_i + \frac{\partial H}{\partial q_i} = 0 \implies \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

These derived relations are Hamilton's canonical equations of motion.

4. Integrals of the Motion, Cyclic Coordinates and Phase Space

Cyclic Coordinates and Conservation of Conjugate Momentum

A generalized coordinate q_k is defined as cyclic (or ignorable) if it does not explicitly appear in the Hamiltonian H, even though its time derivative $\dot{q}_k$ or conjugate momentum p_k may appear. Mathematically, if q_k is cyclic, the partial derivative of H with respect to q_k vanishes:

$$\frac{\partial H}{\partial q_k} = 0$$

Using Hamilton's canonical equation for the time rate of change of the momentum conjugate to q_k:

$$\dot{p}_k = -\frac{\partial H}{\partial q_k}$$

Substituting $\frac{\partial H}{\partial q_k} = 0$ into Hamilton's equation yields

$$\dot{p}_k = 0$$

Integrating this equation with respect to time gives

$$p_k = \text{constant}$$

Thus, whenever a generalized coordinate is cyclic in the Hamiltonian formulation, its corresponding conjugate momentum is a constant of motion.

The Hamiltonian as a Constant of Motion

Consider a dynamic system described by the Hamiltonian $H(q_1, \ldots, q_n, p_1, \ldots, p_n, t)$. The total time derivative of H along a dynamical trajectory is obtained by applying the chain rule of differentiation:

$$\frac{dH}{dt} = \sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) + \frac{\partial H}{\partial t}$$

We substitute Hamilton's equations of motion, $\dot{q}_i = \frac{\partial H}{\partial p_i}$ and $\dot{p}_i = -\frac{\partial H}{\partial q_i}$, into this derivative:

$$\frac{dH}{dt} = \sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} + \frac{\partial H}{\partial p_i} \left(-\frac{\partial H}{\partial q_i} \right) \right) + \frac{\partial H}{\partial t}$$

Combining the products within the summation yields

$$\sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = 0$$

Thus, the total time derivative simplifies directly to

$$\frac{dH}{dt} = \frac{\partial H}{\partial t}$$

If the Hamiltonian does not depend explicitly on time, the partial derivative $\frac{\partial H}{\partial t}$ is zero:

$$\frac{\partial H}{\partial t} = 0$$

Under this condition, the total derivative vanishes:

$$\frac{dH}{dt} = 0 \implies H = \text{constant}$$

Therefore, if the Hamiltonian has no explicit time dependence, H is a constant of motion, which corresponds to the conservation of total mechanical energy for scleronomic conservative systems.

Phase Space and Phase Trajectories

In Hamiltonian mechanics, the state of a physical system with n degrees of freedom is completely specified by n generalized coordinates $q_1, \ldots, q_n$ and n conjugate momenta $p_1, \ldots, p_n$. The mathematical space constructed with these $2n$ independent variables as axes is called the phase space.

A single point in this $2n$-dimensional phase space represents the complete microstate of the system at an instant in time t. As time progresses, the point moves through the phase space according to Hamilton's equations of motion, tracing out a continuous curve called a phase trajectory.

When time t is included explicitly as an independent axis alongside coordinates and momenta, the resulting space is called the extended phase space, which has $2n + 1$ dimensions with coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n, t)$. In extended phase space, phase trajectories corresponding to deterministic dynamic motion never intersect each other.

Constants of Motion in General and Free-Particle Example

A function $f(q_1, \ldots, q_n, p_1, \ldots, p_n, t)$ is called a constant of motion (or an integral of motion) if its value remains invariant along the dynamical path executed by the system. The condition for f to be a constant of motion is that its total time derivative is identically zero:

$$\frac{df}{dt} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right) + \frac{\partial f}{\partial t} = 0$$

As an application of these concepts, consider a free particle of mass m moving in one dimension along the coordinate x with momentum p. The Lagrangian of the free particle is

$$L = \frac{1}{2} m \dot{x}^2$$

The canonical momentum conjugate to x is defined by

$$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

Solving for velocity gives

$$\dot{x} = \frac{p}{m}$$

The Hamiltonian H is constructed via the Legendre transformation:

$$H = p\dot{x} - L = p\left(\frac{p}{m}\right) - \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m}$$

Hamilton's equations for the free particle are

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}$$

$$\dot{p} = -\frac{\partial H}{\partial x} = 0$$

Since the coordinate x does not appear in H, x is cyclic. Integrating $\dot{p} = 0$ gives the first constant of motion:

$$p = p_0 = \text{constant}$$

Since $H = \frac{p^2}{2m}$ has no explicit time dependence ($\frac{\partial H}{\partial t} = 0$), H itself is a constant of motion representing conserved kinetic energy:

$$H = E = \frac{p_0^2}{2m} = \text{constant}$$

Integrating the equation of motion for position $\dot{x} = \frac{p_0}{m}$ with respect to time gives

$$x(t) = \frac{p_0}{m}t + x_0$$

Rearranging this result to isolate the integration constant x_0 yields a time-dependent function:

$$f(x, p, t) = x - \frac{p}{m}t$$

We verify that $f(x, p, t)$ is a constant of motion by taking its total time derivative:

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\dot{x} + \frac{\partial f}{\partial p}\dot{p} + \frac{\partial f}{\partial t}$$

Evaluating the partial derivatives of $f(x, p, t) = x - \frac{p}{m}t$:

$$\frac{\partial f}{\partial x} = 1, \quad \frac{\partial f}{\partial p} = -\frac{t}{m}, \quad \frac{\partial f}{\partial t} = -\frac{p}{m}$$

Substituting these partial derivatives along with $\dot{x} = \frac{p}{m}$ and $\dot{p} = 0$ yields

$$\frac{df}{dt} = (1)\left(\frac{p}{m}\right) + \left(-\frac{t}{m}\right)(0) + \left(-\frac{p}{m}\right) = \frac{p}{m} - \frac{p}{m} = 0$$

Thus, $f(x, p, t) = x - \frac{p}{m}t$ is an explicit time-dependent integral of motion for the free particle.

5. Principle of Least Action

The Principle of Least Action (historically associated with Maupertuis) is an alternative variational principle distinct from Hamilton's principle. While Hamilton's principle considers paths where the initial and final times t_1 and t_2 are strictly fixed and the total energy H may vary along varied comparison paths, the Principle of Least Action considers variations in which the total energy H is strictly conserved ($H = E = \text{constant}$) along both the actual and varied paths, while the transit times at the endpoints are allowed to vary.

To distinguish the variations, we write δ for Hamilton's variations (where $\delta t = 0$) and Δ for variations where the end times are varied ($\Delta t \neq 0$). The total variation Δ of a quantity is related to δ by

$$\Delta = \delta + \dot{(\,)}\Delta t$$

The action W used in the Principle of Least Action (also called the abbreviated action) is expressed in Hamiltonian form as

$$W = \int_{t_1}^{t_2} \sum_{i=1}^{n} p_i \dot{q}_i \, dt$$

Using $\dot{q}_i dt = dq_i$, the abbreviated action can be written independently of explicit time parameterization as

$$W = \int \sum_{i=1}^{n} p_i \, dq_i$$

To see how the Hamiltonian action W relates to the Lagrangian action $I = \int_{t_1}^{t_2} L \, dt$, we recall the relationship between L and H:

$$L = \sum_{i=1}^{n} p_i \dot{q}_i - H$$

Rearranging this gives

$$\sum_{i=1}^{n} p_i \dot{q}_i = L + H$$

Substituting this into the definition of W yields

$$W = \int_{t_1}^{t_2} (L + H) \, dt = \int_{t_1}^{t_2} L \, dt + \int_{t_1}^{t_2} H \, dt$$

Since energy is conserved along the paths under consideration, $H = E$ is a constant. Bringing E outside the time integral gives

$$W = \int_{t_1}^{t_2} L \, dt + E(t_2 - t_1)$$

Thus, the Hamiltonian form of the action W differs from the Lagrangian action integral $I = \int L dt$ by the term $E(t_2 - t_1)$, which accounts for variations in the endpoint times.

The Principle of Least Action states that for dynamic trajectories of a system with conserved energy $H = E$, the variation of the abbreviated action W vanishes for variations where endpoint positions are fixed ($\Delta q_i(t_1) = 0$ and $\Delta q_i(t_2) = 0$):

$$\Delta W = \Delta \int_{t_1}^{t_2} \sum_{i=1}^{n} p_i \dot{q}_i \, dt = 0$$

Although named the Principle of "Least" Action for historical reasons, the condition $\Delta W = 0$ strictly means that the abbreviated action integral is stationary (an extremum, which may be a minimum, maximum, or inflection point) along

the actual path of motion relative to neighboring physical paths of the same constant energy.

6. Canonical Transformations

In Hamiltonian mechanics, the state of a system with n degrees of freedom is specified by n generalized coordinates $q_1, q_2, \ldots, q_n$ and n generalized momenta $p_1, p_2, \ldots, p_n$. These $2n$ variables define the phase space of the system. The time evolution of the system is governed by Hamilton's equations of motion:

$$\dot{q}_k = \frac{\partial H}{\partial p_k}$$

$$\dot{p}_k = -\frac{\partial H}{\partial q_k}$$

where $H(q, p, t)$ is the Hamiltonian of the system and $k = 1, 2, \ldots, n$.

It is often advantageous to transform from the old set of phase space coordinates (q_k, p_k) to a new set of coordinates (Q_k, P_k) defined by transformation equations of the form:

$$Q_k = Q_k(q_1, \ldots, q_n, p_1, \ldots, p_n, t)$$

$$P_k = P_k(q_1, \ldots, q_n, p_1, \ldots, p_n, t)$$

A transformation is defined to be a *canonical transformation* if there exists a new Hamiltonian function $K(Q, P, t)$ such that the equations of motion in the new variables retain the canonical Hamiltonian form:

$$\dot{Q}_k = \frac{\partial K}{\partial P_k}$$

$$\dot{P}_k = -\frac{\partial K}{\partial Q_k}$$

The exact form of Hamilton's equations must be preserved under the transformation for any physical system.

The condition for a transformation to be canonical can be derived from Hamilton's principle of least action. In terms of the original canonical coordinates (q_k, p_k), Hamilton's principle states that the variation of the action integral vanishes:

$$\delta \int_{t_1}^{t_2} \left(\sum_{k=1}^{n} p_k \dot{q}_k - H(q, p, t) \right) dt = 0$$

For the new variables (Q_k, P_k) to obey Hamilton's equations of motion with the new Hamiltonian $K(Q, P, t)$, Hamilton's principle must simultaneously hold in the new coordinate system:

$$\delta \int_{t_1}^{t_2} \left(\sum_{k=1}^{n} P_k \dot{Q}_k - K(Q, P, t) \right) dt = 0$$

The two variational statements are satisfied simultaneously if the integrands differ by at most the total time derivative of an arbitrary function F, called the *generating function*:

$$\sum_{k=1}^{n} p_k \dot{q}_k - H = \sum_{k=1}^{n} P_k \dot{Q}_k - K + \frac{dF}{dt}$$

Multiplying this equation by dt yields the exact differential relation connecting the old and new phase space variables:

$$\sum_{k=1}^{n} p_k dq_k - H dt = \sum_{k=1}^{n} P_k dQ_k - K dt + dF$$

Rearranging the terms to isolate the differential dF gives:

$$dF = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt$$

An alternative and equivalent test for a transformation to be canonical relies on the preservation of fundamental Poisson brackets. A phase space coordinate transformation from (q, p) to (Q, P) is canonical if and only if the new coordinates satisfy the fundamental Poisson bracket relations evaluated with respect to the old variables:

$$\{Q_i, Q_j\}_{q,p} = 0$$

$$\{P_i, P_j\}_{q,p} = 0$$

$$\{Q_i, P_j\}_{q,p} = \delta_{ij}$$

where δ_{ij} is the Kronecker delta, and the Poisson bracket of any two functions $A(q, p)$ and $B(q, p)$ is defined as:

$$\{A, B\}_{q,p} = \sum_{k=1}^{n} \left(\frac{\partial A}{\partial q_k} \frac{\partial B}{\partial p_k} - \frac{\partial A}{\partial p_k} \frac{\partial B}{\partial q_k} \right)$$

If these fundamental bracket relations are satisfied, the transformation preserves the canonical structure of phase space, guaranteeing that Hamilton's equations retain their standard form.

7. Generating Functions

The generating function F serves as the bridge between the old coordinates (q, p) and the new coordinates (Q, P). Because F is a function of phase space variables, it can depend on a mixture of one set of old coordinates and one set of new coordinates, in addition to time t. This leads to four fundamental types of generating functions depending on the choice of independent variables.

Generating Function of the First Kind: $F_1(q, Q, t)$

When the generating function depends on the old generalized coordinates q and the new generalized coordinates Q, we denote it as $F = F_1(q, Q, t)$. Taking the total differential of $F_1(q, Q, t)$ using the chain rule yields:

$$dF_1 = \sum_{k=1}^{n} \frac{\partial F_1}{\partial q_k} dq_k + \sum_{k=1}^{n} \frac{\partial F_1}{\partial Q_k} dQ_k + \frac{\partial F_1}{\partial t} dt$$

We compare this differential with the general condition derived from Hamilton's principle:

$$dF = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt$$

Equating the differential coefficients of the independent variations dq_k, dQ_k, and dt gives the transformation equations for a Type 1 generating function:

$$p_k = \frac{\partial F_1}{\partial q_k}$$

$$P_k = -\frac{\partial F_1}{\partial Q_k}$$

$$K = H + \frac{\partial F_1}{\partial t}$$

Generating Function of the Second Kind: $F_2(q, P, t)$

To obtain a generating function that depends on the old coordinates q and the new momenta P, we perform a Legendre transformation on F_1 to replace Q with P. We define the Type 2 generating function $F_2(q, P, t)$ as:

$$F_2(q, P, t) = F_1(q, Q, t) + \sum_{k=1}^{n} P_k Q_k$$

Taking the total differential of both sides of this definition gives:

$$dF_2 = dF_1 + \sum_{k=1}^{n} P_k dQ_k + \sum_{k=1}^{n} Q_k dP_k$$

We substitute the fundamental condition for dF_1:

$$dF_1 = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt$$

into the differential expression for dF_2:

$$dF_2 = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt + \sum_{k=1}^{n} P_k dQ_k + \sum_{k=1}^{n} Q_k dP_k$$

Canceling the matching terms $\sum_{k=1}^{n} P_k dQ_k$ simplifies the expression to:

$$dF_2 = \sum_{k=1}^{n} p_k dq_k + \sum_{k=1}^{n} Q_k dP_k + (K - H)dt$$

Expanding $dF_2(q, P, t)$ directly in terms of its partial derivatives yields:

$$dF_2 = \sum_{k=1}^{n} \frac{\partial F_2}{\partial q_k} dq_k + \sum_{k=1}^{n} \frac{\partial F_2}{\partial P_k} dP_k + \frac{\partial F_2}{\partial t} dt$$

Equating the coefficients of dq_k, dP_k, and dt from the two differential forms gives the transformation equations:

$$p_k = \frac{\partial F_2}{\partial q_k}$$

$$Q_k = \frac{\partial F_2}{\partial P_k}$$

$$K = H + \frac{\partial F_2}{\partial t}$$

Generating Function of the Third Kind: $F_3(p, Q, t)$

To construct a generating function that depends on the old momenta p and the new coordinates Q, we perform a Legendre transformation on F_1 to replace q with p. We define the Type 3 generating function $F_3(p, Q, t)$ as:

$$F_3(p, Q, t) = F_1(q, Q, t) - \sum_{k=1}^{n} p_k q_k$$

Taking the total differential of both sides of this definition gives:

$$dF_3 = dF_1 - \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} q_k dp_k$$

We substitute $dF_1 = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt$ into this differential:

$$dF_3 = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt - \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} q_k dp_k$$

Canceling the matching terms $\sum_{k=1}^{n} p_k dq_k$ simplifies the expression to:

$$dF_3 = -\sum_{k=1}^{n} q_k dp_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt$$

Expanding $dF_3(p, Q, t)$ directly in terms of its partial derivatives yields:

$$dF_3 = \sum_{k=1}^{n} \frac{\partial F_3}{\partial p_k} dp_k + \sum_{k=1}^{n} \frac{\partial F_3}{\partial Q_k} dQ_k + \frac{\partial F_3}{\partial t} dt$$

Equating the coefficients of dp_k, dQ_k, and dt from the two differential forms gives the transformation equations:

$$q_k = -\frac{\partial F_3}{\partial p_k}$$

$$P_k = -\frac{\partial F_3}{\partial Q_k}$$

$$K = H + \frac{\partial F_3}{\partial t}$$

Generating Function of the Fourth Kind: $F_4(p, P, t)$

To construct a generating function depending on the old momenta p and the new momenta P, we perform a double Legendre transformation on F_1 to replace both q and Q. We define the Type 4 generating function $F_4(p, P, t)$ as:

$$F_4(p, P, t) = F_1(q, Q, t) - \sum_{k=1}^{n} p_k q_k + \sum_{k=1}^{n} P_k Q_k$$

Taking the total differential of both sides of this definition gives:

$$dF_4 = dF_1 - \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} q_k dp_k + \sum_{k=1}^{n} P_k dQ_k + \sum_{k=1}^{n} Q_k dP_k$$

We substitute $dF_1 = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt$ into this differential:

$$dF_4 = \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} P_k dQ_k + (K - H)dt - \sum_{k=1}^{n} p_k dq_k - \sum_{k=1}^{n} q_k dp_k + \sum_{k=1}^{n} P_k dQ_k + \sum_{k=1}^{n} Q_k dP_k$$

Canceling the matching terms $\sum_{k=1}^{n} p_k dq_k$ and $\sum_{k=1}^{n} P_k dQ_k$ simplifies the expression to:

$$dF_4 = -\sum_{k=1}^{n} q_k dp_k + \sum_{k=1}^{n} Q_k dP_k + (K - H)dt$$

Expanding $dF_4(p, P, t)$ directly in terms of its partial derivatives yields:

$$dF_4 = \sum_{k=1}^{n} \frac{\partial F_4}{\partial p_k} dp_k + \sum_{k=1}^{n} \frac{\partial F_4}{\partial P_k} dP_k + \frac{\partial F_4}{\partial t} dt$$

Equating the coefficients of dp_k, dP_k, and dt from the two differential forms gives the transformation equations:

$$q_k = -\frac{\partial F_4}{\partial p_k}$$

$$Q_k = \frac{\partial F_4}{\partial P_k}$$

$$K = H + \frac{\partial F_4}{\partial t}$$

8. Worked Example: A Canonical Transformation

Consider a system with one degree of freedom described by the original canonical coordinates (q, p). We examine the coordinate transformation defined by:

$$P = \frac{1}{2}(p^2 + q^2)$$

$$Q = \tan^{-1}\left(\frac{q}{p}\right)$$

We will demonstrate that this transformation is canonical using two distinct methods: first, by calculating the fundamental Poisson bracket $\{Q, P\}_{q,p}$, and second, by constructing an explicit generating function.

Method 1: Verification via Fundamental Poisson Bracket

For a system with one degree of freedom, the Poisson bracket of Q and P with respect to (q, p) is defined as:

$$\{Q, P\} = \frac{\partial Q}{\partial q}\frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial P}{\partial q}$$

We evaluate each partial derivative individually.

First, differentiating $Q = \tan^{-1}(q/p)$ with respect to q using the chain rule gives:

$$\frac{\partial Q}{\partial q} = \frac{1}{1 + (q/p)^2} \cdot \frac{\partial}{\partial q}\left(\frac{q}{p}\right)$$

Evaluating the inner derivative yields:

$$\frac{\partial Q}{\partial q} = \frac{1}{1 + q^2/p^2} \cdot \frac{1}{p}$$

Multiplying the numerator and denominator by p^2 simplifies the expression:

$$\frac{\partial Q}{\partial q} = \frac{p^2}{p^2 + q^2} \cdot \frac{1}{p}$$

$$\frac{\partial Q}{\partial q} = \frac{p}{p^2 + q^2}$$

Second, differentiating $Q = \tan^{-1}(q/p)$ with respect to p gives:

$$\frac{\partial Q}{\partial p} = \frac{1}{1 + (q/p)^2} \cdot \frac{\partial}{\partial p}\left(\frac{q}{p}\right)$$

Evaluating the inner derivative yields:

$$\frac{\partial Q}{\partial p} = \frac{1}{1 + q^2/p^2} \cdot \left(-\frac{q}{p^2}\right)$$

Multiplying the numerator and denominator by p^2 simplifies the expression:

$$\frac{\partial Q}{\partial p} = \frac{p^2}{p^2 + q^2} \cdot \left(-\frac{q}{p^2}\right)$$

$$\frac{\partial Q}{\partial p} = -\frac{q}{p^2 + q^2}$$

Third, differentiating $P = \frac{1}{2}(p^2 + q^2)$ with respect to q yields:

$$\frac{\partial P}{\partial q} = q$$

Fourth, differentiating $P = \frac{1}{2}(p^2 + q^2)$ with respect to p yields:

$$\frac{\partial P}{\partial p} = p$$

Now, substituting these four partial derivatives into the Poisson bracket formula gives:

$$\{Q, P\} = \left(\frac{p}{p^2 + q^2}\right)(p) - \left(-\frac{q}{p^2 + q^2}\right)(q)$$

Multiplying out the terms in the expression gives:

$$\{Q, P\} = \frac{p^2}{p^2+q^2} + \frac{q^2}{p^2+q^2}$$

Combining the fractions over the common denominator gives:

$$\{Q, P\} = \frac{p^2+q^2}{p^2+q^2}$$

$$\{Q, P\} = 1$$

Since $\{Q, P\} = 1$, the fundamental Poisson bracket is preserved. Therefore, the transformation is canonical.

Method 2: Verification via Generating Function

We seek a generating function of the first kind, $F_1(q, Q)$, to generate this canonical transformation. Taking the tangent of both sides of the transformation equation for Q:

$$\tan Q = \frac{q}{p}$$

Solving this relation for the momentum p in terms of q and Q yields:

$$p = \frac{q}{\tan Q}$$

$$p = q \cot Q$$

The transformation relation for a Type 1 generating function $F_1(q, Q)$ specifies that:

$$p = \frac{\partial F_1}{\partial q}$$

Equating this general relation to our expression for p gives:

$$\frac{\partial F_1}{\partial q} = q \cot Q$$

Integrating both sides with respect to q treating Q as constant yields:

$$F_1(q, Q) = \int q \cot Q \, dq$$

$$F_1(q, Q) = \frac{1}{2} q^2 \cot Q + C(Q)$$

where $C(Q)$ is an integration constant with respect to q. Choosing $C(Q) = 0$ gives the generating function:

$$F_1(q, Q) = \frac{1}{2} q^2 \cot Q$$

We must now verify that this generating function correctly reproduces the transformation rule for P. The Type 1 relation for P is:

$$P = -\frac{\partial F_1}{\partial Q}$$

Differentiating $F_1(q, Q) = \frac{1}{2}q^2 \cot Q$ with respect to Q yields:

$$P = -\frac{\partial}{\partial Q}\left(\frac{1}{2}q^2 \cot Q\right)$$

Using the trigonometric derivative $\frac{d}{dQ}(\cot Q) = -\csc^2 Q$:

$$P = -\frac{1}{2}q^2(-\csc^2 Q)$$

$$P = \frac{1}{2}q^2 \csc^2 Q$$

We apply the trigonometric identity $\csc^2 Q = 1 + \cot^2 Q$:

$$P = \frac{1}{2}q^2(1 + \cot^2 Q)$$

Substituting $\cot Q = p/q$ into this expression gives:

$$P = \frac{1}{2}q^2\left(1 + \frac{p^2}{q^2}\right)$$

Distributing q^2 through the terms in parentheses gives:

$$P = \frac{1}{2}(q^2 + p^2)$$

$$P = \frac{1}{2}(p^2 + q^2)$$

This matches the given definition of P. Because an explicit generating function $F_1(q, Q) = \frac{1}{2}q^2 \cot Q$ exists and generates both transformation equations, the transformation is proved to be canonical.

9. Infinitesimal Canonical Transformations

An infinitesimal canonical transformation is a canonical transformation in which the new canonical variables (Q_j, P_j) differ from the old canonical variables (q_j, p_j) by infinitesimal quantities. Such a transformation can be generated by a type-2 generating function $F_2(q, P, t)$ that differs infinitesimally from the identity transformation generator $\sum_{j=1}^{n} q_j P_j$.

We define the generating function for an infinitesimal canonical transformation as

$$F_2(q, P, t) = \sum_{j=1}^{n} q_j P_j + \epsilon G(q, P, t)$$

where ϵ is an infinitesimal parameter and $G(q, P, t)$ is a differentiable function of the coordinates, momenta, and time, referred to as the generator of the infinitesimal transformation.

The general canonical transformation equations derived from $F_2(q, P, t)$ are

$$p_j = \frac{\partial F_2}{\partial q_j}$$

$$Q_j = \frac{\partial F_2}{\partial P_j}$$

Substituting the explicit form of $F_2(q, P, t)$ into the equation for the old momentum p_j, we obtain

$$p_j = \frac{\partial}{\partial q_j}\left(\sum_{k=1}^{n} q_k P_k + \epsilon G(q, P, t)\right)$$

Carrying out the partial derivative yields

$$p_j = P_j + \epsilon \frac{\partial G(q, P, t)}{\partial q_j}$$

Similarly, substituting $F_2(q, P, t)$ into the equation for the new coordinate Q_j, we get

$$Q_j = \frac{\partial}{\partial P_j}\left(\sum_{k=1}^{n} q_k P_k + \epsilon G(q, P, t)\right)$$

Carrying out the partial derivative yields

$$Q_j = q_j + \epsilon \frac{\partial G(q, P, t)}{\partial P_j}$$

We define the infinitesimal changes in the coordinate q_j and momentum p_j as

$$\delta q_j = Q_j - q_j$$

$$\delta p_j = P_j - p_j$$

Rearranging the transformation equations gives

$$\delta q_j = \epsilon \frac{\partial G(q, P, t)}{\partial P_j}$$

$$\delta p_j = -\epsilon \frac{\partial G(q, P, t)}{\partial q_j}$$

Since ϵ is an infinitesimal parameter, the difference between P_j and p_j is of order ϵ. Therefore, evaluating $G(q, P, t)$ at $P_j = p_j$ introduces an error of order ϵ^2, which is negligible. To first order in ϵ, we replace P_j with p_j in the generator G, writing it as $G(q, p, t)$.

The infinitesimal changes in the canonical variables expressed in terms of $G(q, p, t)$ become

$$\delta q_j = \epsilon \frac{\partial G(q, p, t)}{\partial p_j}$$

$$\delta p_j = -\epsilon \frac{\partial G(q, p, t)}{\partial q_j}$$

Using the definition of the Poisson bracket between two functions, these infinitesimal transformations can be expressed concisely as

$$\delta q_j = \epsilon\{q_j, G\}$$

$$\delta p_j = \epsilon\{p_j, G\}$$

Next, we evaluate the infinitesimal change δu in an arbitrary dynamical variable $u(q, p, t)$ under an infinitesimal canonical transformation. The change δu is defined as

$$\delta u = u(Q, P, t) - u(q, p, t)$$

Expanding $u(Q, P, t) = u(q + \delta q, p + \delta p, t)$ to first order in the infinitesimal variations δq_j and δp_j via Taylor series gives

$$\delta u = \sum_{j=1}^{n} \left(\frac{\partial u}{\partial q_j} \delta q_j + \frac{\partial u}{\partial p_j} \delta p_j \right)$$

Substituting the expressions for $\delta q_j = \epsilon \frac{\partial G}{\partial p_j}$ and $\delta p_j = -\epsilon \frac{\partial G}{\partial q_j}$ into this expansion yields

$$\delta u = \sum_{j=1}^{n} \left(\frac{\partial u}{\partial q_j} \left(\epsilon \frac{\partial G}{\partial p_j} \right) + \frac{\partial u}{\partial p_j} \left(-\epsilon \frac{\partial G}{\partial q_j} \right) \right)$$

Factoring out the infinitesimal parameter ϵ, we obtain

$$\delta u = \epsilon \sum_{j=1}^{n} \left(\frac{\partial u}{\partial q_j} \frac{\partial G}{\partial p_j} - \frac{\partial u}{\partial p_j} \frac{\partial G}{\partial q_j} \right)$$

Recognizing the sum as the Poisson bracket of u and G, we arrive at

$$\delta u = \epsilon\{u, G\}$$

An essential application of infinitesimal canonical transformations is the representation of continuous time evolution. Consider the Hamiltonian $H(q, p, t)$ of a system as the generator of an infinitesimal canonical transformation, with the infinitesimal parameter chosen as the time increment $\epsilon = dt$.

Substituting $G = H$ and $\epsilon = dt$ into the infinitesimal transformation equations for coordinates and momenta gives

$$\delta q_j = dt\{q_j, H\}$$

$$\delta p_j = dt\{p_j, H\}$$

Using Hamilton's equations of motion, $\dot{q}_j = \frac{\partial H}{\partial p_j} = \{q_j, H\}$ and $\dot{p}_j = -\frac{\partial H}{\partial q_j} = \{p_j, H\}$, we rewrite the variations as

$$\delta q_j = \dot{q}_j dt = dq_j$$

$$\delta p_j = \dot{p}_j dt = dp_j$$

Thus, the actual physical motion of a Hamiltonian system from time t to time $t + dt$ is mathematically equivalent to an infinitesimal canonical transformation generated by the Hamiltonian H.

10. Poisson Brackets

The Poisson bracket provides a fundamental algebraic structure in classical mechanics that connects dynamical variables with the equations of motion and conservation laws.

Given two dynamic variables $f(q,p,t)$ and $g(q,p,t)$ defined on the phase space of a system with n degrees of freedom, the Poisson bracket of f and g with respect to the canonical variables $(q,p) = (q_1,\dots,q_n,p_1,\dots,p_n)$ is defined as

$$\{f,g\}_{q,p} = \sum_{j=1}^{n}\left(\frac{\partial f}{\partial q_j}\frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_j}\frac{\partial g}{\partial q_j}\right)$$

When the choice of canonical variables is clear from the context, the subscript (q,p) is usually omitted.

The fundamental Poisson brackets are the Poisson brackets evaluated directly between the canonical coordinates q_j and canonical momenta p_k.

First, we evaluate $\{q_j,q_k\}$ using the definition of the Poisson bracket:

$$\{q_j,q_k\} = \sum_{l=1}^{n}\left(\frac{\partial q_j}{\partial q_l}\frac{\partial q_k}{\partial p_l} - \frac{\partial q_j}{\partial p_l}\frac{\partial q_k}{\partial q_l}\right)$$

Since canonical coordinates and canonical momenta are independent variables, $\frac{\partial q_j}{\partial p_l} = 0$ for all j,l. Thus, both terms in the sum vanish:

$$\{q_j,q_k\} = 0$$

Second, we evaluate $\{p_j,p_k\}$ using the definition:

$$\{p_j,p_k\} = \sum_{l=1}^{n}\left(\frac{\partial p_j}{\partial q_l}\frac{\partial p_k}{\partial p_l} - \frac{\partial p_j}{\partial p_l}\frac{\partial p_k}{\partial q_l}\right)$$

Since $\frac{\partial p_j}{\partial q_l} = 0$ for all j,l, both terms in the sum vanish:

$$\{p_j,p_k\} = 0$$

Third, we evaluate $\{q_j,p_k\}$ using the definition:

$$\{q_j,p_k\} = \sum_{l=1}^{n}\left(\frac{\partial q_j}{\partial q_l}\frac{\partial p_k}{\partial p_l} - \frac{\partial q_j}{\partial p_l}\frac{\partial p_k}{\partial q_l}\right)$$

Using the partial derivative properties $\frac{\partial q_j}{\partial q_l} = \delta_{jl}$, $\frac{\partial p_k}{\partial p_l} = \delta_{kl}$, and $\frac{\partial q_j}{\partial p_l} = 0$, the sum simplifies to

$$\{q_j,p_k\} = \sum_{l=1}^{n}\delta_{jl}\delta_{kl}$$

Evaluating the summation over l gives

$$\{q_j,p_k\} = \delta_{jk}$$

where δ_{jk} is the Kronecker delta, equal to 1 if $j=k$ and 0 if $j\neq k$.

Summarizing, the fundamental Poisson brackets are

$$\{q_j, q_k\} = 0$$

$$\{p_j, p_k\} = 0$$

$$\{q_j, p_k\} = \delta_{jk}$$

11. Fundamental Properties of Poisson Brackets

Poisson brackets satisfy several core algebraic properties: antisymmetry, linearity, the Leibniz product rule, and the Jacobi identity.

Antisymmetry

For any two dynamic variables f and g, the Poisson bracket is antisymmetric under the exchange of f and g:

$$\{f, g\} = -\{g, f\}$$

To prove this, we write out the definition of $\{f, g\}$:

$$\{f, g\} = \sum_{j=1}^{n} \left(\frac{\partial f}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial g}{\partial q_j} \right)$$

Factoring out a negative sign from each term in the sum gives

$$\{f, g\} = -\sum_{j=1}^{n} \left(\frac{\partial g}{\partial q_j} \frac{\partial f}{\partial p_j} - \frac{\partial g}{\partial p_j} \frac{\partial f}{\partial q_j} \right)$$

Recognizing the sum as the definition of $\{g, f\}$, we obtain

$$\{f, g\} = -\{g, f\}$$

Setting $g = f$ immediately yields $\{f, f\} = 0$.

Linearity

The Poisson bracket is linear with respect to constant linear combinations of functions:

$$\{c_1 f_1 + c_2 f_2, g\} = c_1\{f_1, g\} + c_2\{f_2, g\}$$

where c_1 and c_2 are constants.

To prove linearity, we apply the definition of the Poisson bracket to the function $(c_1 f_1 + c_2 f_2)$:

$$\{c_1 f_1 + c_2 f_2, g\} = \sum_{j=1}^{n} \left(\frac{\partial (c_1 f_1 + c_2 f_2)}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial (c_1 f_1 + c_2 f_2)}{\partial p_j} \frac{\partial g}{\partial q_j} \right)$$

Using the linearity of partial differentiation, we expand the terms:

$$\{c_1 f_1 + c_2 f_2, g\} = \sum_{j=1}^{n} \left(\left(c_1 \frac{\partial f_1}{\partial q_j} + c_2 \frac{\partial f_2}{\partial q_j} \right) \frac{\partial g}{\partial p_j} - \left(c_1 \frac{\partial f_1}{\partial p_j} + c_2 \frac{\partial f_2}{\partial p_j} \right) \frac{\partial g}{\partial q_j} \right)$$

Distributing the products and regrouping terms proportional to c_1 and c_2 gives

$$\{c_1 f_1 + c_2 f_2, g\} = c_1 \sum_{j=1}^{n} \left(\frac{\partial f_1}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f_1}{\partial p_j} \frac{\partial g}{\partial q_j} \right) + c_2 \sum_{j=1}^{n} \left(\frac{\partial f_2}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f_2}{\partial p_j} \frac{\partial g}{\partial q_j} \right)$$

Expressing both sums in terms of Poisson brackets, we obtain

$$\{c_1 f_1 + c_2 f_2, g\} = c_1 \{f_1, g\} + c_2 \{f_2, g\}$$

Leibniz Rule (Product Rule)

For any three functions f, g, and h, the Poisson bracket satisfies the product rule:

$$\{fg, h\} = f\{g, h\} + \{f, h\}g$$

To derive this property, we start from the definition of $\{fg, h\}$:

$$\{fg, h\} = \sum_{j=1}^{n} \left(\frac{\partial (fg)}{\partial q_j} \frac{\partial h}{\partial p_j} - \frac{\partial (fg)}{\partial p_j} \frac{\partial h}{\partial q_j} \right)$$

Applying the ordinary product rule for partial differentiation to $\frac{\partial (fg)}{\partial q_j}$ and $\frac{\partial (fg)}{\partial p_j}$, we expand the expression as

$$\{fg, h\} = \sum_{j=1}^{n} \left(\left(f \frac{\partial g}{\partial q_j} + g \frac{\partial f}{\partial q_j} \right) \frac{\partial h}{\partial p_j} - \left(f \frac{\partial g}{\partial p_j} + g \frac{\partial f}{\partial p_j} \right) \frac{\partial h}{\partial q_j} \right)$$

Rearranging terms into two separate sums yields

$$\{fg, h\} = f \sum_{j=1}^{n} \left(\frac{\partial g}{\partial q_j} \frac{\partial h}{\partial p_j} - \frac{\partial g}{\partial p_j} \frac{\partial h}{\partial q_j} \right) + g \sum_{j=1}^{n} \left(\frac{\partial f}{\partial q_j} \frac{\partial h}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial h}{\partial q_j} \right)$$

Replacing the sums with their corresponding Poisson bracket definitions gives

$$\{fg, h\} = f\{g, h\} + \{f, h\}g$$

Jacobi Identity

For any three arbitrary functions f, g, and h, the Poisson bracket satisfies the Jacobi identity:

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$$

To prove the identity, we examine the second-order partial derivative terms generated by each double bracket term.

Consider the first term, $\{f, \{g, h\}\}$:

$$\{f, \{g, h\}\} = \sum_{j=1}^{n} \left(\frac{\partial f}{\partial q_j} \frac{\partial \{g, h\}}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial \{g, h\}}{\partial q_j} \right)$$

Substituting $\{g,h\} = \sum_{k=1}^{n}\left(\frac{\partial g}{\partial q_k}\frac{\partial h}{\partial p_k} - \frac{\partial g}{\partial p_k}\frac{\partial h}{\partial q_k}\right)$ into this expression yields terms containing second derivatives of g and h, multiplied by first derivatives of f.

Specifically, the second-derivative terms in $\{f,\{g,h\}\}$ take the form:

$$\sum_{j,k}\frac{\partial f}{\partial q_j}\left(\frac{\partial^2 g}{\partial p_j \partial q_k}\frac{\partial h}{\partial p_k} + \frac{\partial g}{\partial q_k}\frac{\partial^2 h}{\partial p_j \partial p_k} - \frac{\partial^2 g}{\partial p_j \partial p_k}\frac{\partial h}{\partial q_k} - \frac{\partial g}{\partial p_k}\frac{\partial^2 h}{\partial p_j \partial q_k}\right)$$

$$-\sum_{j,k}\frac{\partial f}{\partial p_j}\left(\frac{\partial^2 g}{\partial q_j \partial q_k}\frac{\partial h}{\partial p_k} + \frac{\partial g}{\partial q_k}\frac{\partial^2 h}{\partial q_j \partial p_k} - \frac{\partial^2 g}{\partial q_j \partial p_k}\frac{\partial h}{\partial q_k} - \frac{\partial g}{\partial p_k}\frac{\partial^2 h}{\partial q_j \partial q_k}\right)$$

Evaluating the remaining two terms of the Jacobi identity, $\{g,\{h,f\}\}$ and $\{h,\{f,g\}\}$, produces analogous expansions with cyclic permutations of (f,g,h).

Summing all three double Poisson brackets causes every second-derivative term to cancel in pairs due to the equality of mixed partial derivatives (e.g., $\frac{\partial^2 g}{\partial q_j \partial q_k} = \frac{\partial^2 g}{\partial q_k \partial q_j}$).

Thus, the total sum vanishes identically:

$$\{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\} = 0$$

12. Equations of Motion in Poisson Bracket Form

The time evolution of any dynamical variable $f(q_1,\ldots,q_n,p_1,\ldots,p_n,t)$ along a trajectory in phase space is governed by the total time derivative $\frac{df}{dt}$.

Using the chain rule of partial differentiation, the total time derivative of f is written as

$$\frac{df}{dt} = \sum_{j=1}^{n}\left(\frac{\partial f}{\partial q_j}\frac{dq_j}{dt} + \frac{\partial f}{\partial p_j}\frac{dp_j}{dt}\right) + \frac{\partial f}{\partial t}$$

Hamilton's equations of motion express the time derivatives of the canonical variables in terms of partial derivatives of the Hamiltonian H:

$$\frac{dq_j}{dt} = \frac{\partial H}{\partial p_j}$$

$$\frac{dp_j}{dt} = -\frac{\partial H}{\partial q_j}$$

Substituting Hamilton's equations into the total time derivative expansion gives

$$\frac{df}{dt} = \sum_{j=1}^{n}\left(\frac{\partial f}{\partial q_j}\left(\frac{\partial H}{\partial p_j}\right) + \frac{\partial f}{\partial p_j}\left(-\frac{\partial H}{\partial q_j}\right)\right) + \frac{\partial f}{\partial t}$$

Grouping the summation terms together yields

$$\frac{df}{dt} = \sum_{j=1}^{n}\left(\frac{\partial f}{\partial q_j}\frac{\partial H}{\partial p_j} - \frac{\partial f}{\partial p_j}\frac{\partial H}{\partial q_j}\right) + \frac{\partial f}{\partial t}$$

Recognizing the summation as the Poisson bracket $\{f, H\}$, we obtain the general equation of motion for any dynamical variable f:

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}$$

Hamilton's equations of motion themselves can be expressed directly in Poisson bracket form by choosing f to be the fundamental canonical coordinates and momenta.

Setting $f = q_k$, since $\frac{\partial q_k}{\partial t} = 0$, we get

$$\frac{dq_k}{dt} = \{q_k, H\} + 0$$

$$\frac{dq_k}{dt} = \{q_k, H\}$$

Setting $f = p_k$, since $\frac{\partial p_k}{\partial t} = 0$, we get

$$\frac{dp_k}{dt} = \{p_k, H\} + 0$$

$$\frac{dp_k}{dt} = \{p_k, H\}$$

These two equations represent Hamilton's equations of motion expressed entirely in Poisson bracket notation.

A quantity $f(q, p, t)$ is a constant of motion (or an integral of motion) if its total time derivative vanishes along the dynamical trajectory:

$$\frac{df}{dt} = 0$$

From the general equation of motion, the condition for f to be a constant of motion is

$$\{f, H\} + \frac{\partial f}{\partial t} = 0$$

If the dynamical variable f has no explicit dependence on time ($\frac{\partial f}{\partial t} = 0$), the condition simplifies to

$$\{f, H\} = 0$$

Thus, a time-independent dynamical variable is a constant of motion if and only if its Poisson bracket with the Hamiltonian vanishes.

13. Invariance of Poisson Brackets

A defining characteristic of canonical transformations is that they preserve the structure of Poisson brackets. That is, the Poisson bracket of any two functions f and g evaluated with respect to the old variables (q, p) is equal to their Poisson bracket evaluated with respect to the new canonical variables (Q, P):

$$\{f, g\}_{q,p} = \{f, g\}_{Q,P}$$

To prove canonical invariance, we express the partial derivatives of f and g with respect to (q_j, p_j) using the multivariable chain rule in terms of (Q_k, P_k):

$$\frac{\partial f}{\partial q_j} = \sum_{k=1}^{n} \left(\frac{\partial f}{\partial Q_k} \frac{\partial Q_k}{\partial q_j} + \frac{\partial f}{\partial P_k} \frac{\partial P_k}{\partial q_j} \right)$$

$$\frac{\partial f}{\partial p_j} = \sum_{k=1}^{n} \left(\frac{\partial f}{\partial Q_k} \frac{\partial Q_k}{\partial p_j} + \frac{\partial f}{\partial P_k} \frac{\partial P_k}{\partial p_j} \right)$$

Similarly, for the function g:

$$\frac{\partial g}{\partial q_j} = \sum_{l=1}^{n} \left(\frac{\partial g}{\partial Q_l} \frac{\partial Q_l}{\partial q_j} + \frac{\partial g}{\partial P_l} \frac{\partial P_l}{\partial q_j} \right)$$

$$\frac{\partial g}{\partial p_j} = \sum_{l=1}^{n} \left(\frac{\partial g}{\partial Q_l} \frac{\partial Q_l}{\partial p_j} + \frac{\partial g}{\partial P_l} \frac{\partial P_l}{\partial p_j} \right)$$

Substituting these chain rule expansions into the definition of $\{f, g\}_{q,p}$ yields

$$\{f, g\}_{q,p} = \sum_{j=1}^{n} \left[\left(\sum_{k=1}^{n} \left(\frac{\partial f}{\partial Q_k} \frac{\partial Q_k}{\partial q_j} + \frac{\partial f}{\partial P_k} \frac{\partial P_k}{\partial q_j} \right) \right) \left(\sum_{l=1}^{n} \left(\frac{\partial g}{\partial Q_l} \frac{\partial Q_l}{\partial p_j} + \frac{\partial g}{\partial P_l} \frac{\partial P_l}{\partial p_j} \right) \right) \right.$$
$$\left. - \left(\sum_{k=1}^{n} \left(\frac{\partial f}{\partial Q_k} \frac{\partial Q_k}{\partial p_j} + \frac{\partial f}{\partial P_k} \frac{\partial P_k}{\partial p_j} \right) \right) \left(\sum_{l=1}^{n} \left(\frac{\partial g}{\partial Q_l} \frac{\partial Q_l}{\partial q_j} + \frac{\partial g}{\partial P_l} \frac{\partial P_l}{\partial q_j} \right) \right) \right]$$

Expanding the products and regrouping terms according to the partial derivatives of f and g with respect to the new variables (Q_k, P_k, Q_l, P_l), we get

$$\{f, g\}_{q,p} = \sum_{k=1}^{n} \sum_{l=1}^{n} \left[\frac{\partial f}{\partial Q_k} \frac{\partial g}{\partial Q_l} \{Q_k, Q_l\}_{q,p} + \frac{\partial f}{\partial Q_k} \frac{\partial g}{\partial P_l} \{Q_k, P_l\}_{q,p} \right.$$
$$\left. + \frac{\partial f}{\partial P_k} \frac{\partial g}{\partial Q_l} \{P_k, Q_l\}_{q,p} + \frac{\partial f}{\partial P_k} \frac{\partial g}{\partial P_l} \{P_k, P_l\}_{q,p} \right]$$

Since the transformation $(q, p) \to (Q, P)$ is canonical, the fundamental Poisson brackets with respect to (q, p) satisfy

$$\{Q_k, Q_l\}_{q,p} = 0$$

$$\{P_k, P_l\}_{q,p} = 0$$

$$\{Q_k, P_l\}_{q,p} = \delta_{kl}$$

$$\{P_k, Q_l\}_{q,p} = -\delta_{kl}$$

Substituting these fundamental relations into the expanded sum eliminates the first and fourth terms:

$$\{f, g\}_{q,p} = \sum_{k=1}^{n} \sum_{l=1}^{n} \left[\frac{\partial f}{\partial Q_k} \frac{\partial g}{\partial P_l} \delta_{kl} - \frac{\partial f}{\partial P_k} \frac{\partial g}{\partial Q_l} \delta_{kl} \right]$$

Evaluating the summation over l using the Kronecker delta gives

$$\{f,g\}_{q,p} = \sum_{k=1}^{n} \left(\frac{\partial f}{\partial Q_k}\frac{\partial g}{\partial P_k} - \frac{\partial f}{\partial P_k}\frac{\partial g}{\partial Q_k} \right)$$

Recognizing the right-hand side as $\{f,g\}_{Q,P}$, we complete the proof:

$$\{f,g\}_{q,p} = \{f,g\}_{Q,P}$$

We now check this canonical invariance explicitly using the harmonic oscillator transformation. Consider the transformation between old canonical variables (q,p) and new variables (Q,P) given by

$$q = \sqrt{2P}\sin Q$$

$$p = \sqrt{2P}\cos Q$$

To check whether this transformation is canonical, we calculate the Poisson bracket $\{q,p\}_{Q,P}$ with respect to the new variables (Q,P):

$$\{q,p\}_{Q,P} = \frac{\partial q}{\partial Q}\frac{\partial p}{\partial P} - \frac{\partial q}{\partial P}\frac{\partial p}{\partial Q}$$

First, we compute the four partial derivatives of q and p with respect to Q and P:

$$\frac{\partial q}{\partial Q} = \frac{\partial}{\partial Q}(\sqrt{2P}\sin Q) = \sqrt{2P}\cos Q$$

$$\frac{\partial p}{\partial P} = \frac{\partial}{\partial P}(\sqrt{2P}\cos Q) = \frac{1}{\sqrt{2P}}\cos Q$$

$$\frac{\partial q}{\partial P} = \frac{\partial}{\partial P}(\sqrt{2P}\sin Q) = \frac{1}{\sqrt{2P}}\sin Q$$

$$\frac{\partial p}{\partial Q} = \frac{\partial}{\partial Q}(\sqrt{2P}\cos Q) = -\sqrt{2P}\sin Q$$

Next, we substitute these partial derivatives back into the Poisson bracket formula:

$$\{q,p\}_{Q,P} = (\sqrt{2P}\cos Q)\left(\frac{1}{\sqrt{2P}}\cos Q\right) - \left(\frac{1}{\sqrt{2P}}\sin Q\right)(-\sqrt{2P}\sin Q)$$

Simplifying each term in the product gives

$$\{q,p\}_{Q,P} = \cos^2 Q + \sin^2 Q$$

Using the trigonometric identity $\cos^2 Q + \sin^2 Q = 1$, we obtain

$$\{q,p\}_{Q,P} = 1$$

Since $\{q,p\}_{q,p} = 1$ in the old variables, we have verified that $\{q,p\}_{Q,P} = \{q,p\}_{q,p} = 1$, confirming that the transformation is canonical and that the Poisson bracket is invariant.

14. Lagrange Brackets

The Lagrange bracket is a differential structure closely related to the Poisson bracket.

Let u and v be any two variables that are functions of the canonical coordinates and momenta $(q,p) = (q_1,\ldots,q_n,p_1,\ldots,p_n)$. The Lagrange bracket of u and v with respect to (q,p) is defined as

$$[u,v]_{q,p} = \sum_{j=1}^{n}\left(\frac{\partial q_j}{\partial u}\frac{\partial p_j}{\partial v} - \frac{\partial p_j}{\partial u}\frac{\partial q_j}{\partial v}\right)$$

Evaluating the Lagrange brackets directly for the fundamental canonical variables (q,p) yields the fundamental Lagrange brackets:

$$[q_j,q_k] = 0$$

$$[p_j,p_k] = 0$$

$$[q_j,p_k] = \delta_{jk}$$

The Lagrange bracket satisfies key algebraic properties:

1. **Antisymmetry**: Swapping the variables u and v reverses the sign of the bracket:

$$[u,v] = -[v,u]$$

This follows directly from the definition:

$$[u,v] = \sum_{j=1}^{n}\left(\frac{\partial q_j}{\partial u}\frac{\partial p_j}{\partial v} - \frac{\partial p_j}{\partial u}\frac{\partial q_j}{\partial v}\right) = -\sum_{j=1}^{n}\left(\frac{\partial q_j}{\partial v}\frac{\partial p_j}{\partial u} - \frac{\partial p_j}{\partial v}\frac{\partial q_j}{\partial u}\right) = -[v,u]$$

1. **Canonical Invariance**: Like Poisson brackets, Lagrange brackets are invariant under canonical transformations. If $(q,p) \to (Q,P)$ is a canonical transformation, then

$$[u,v]_{q,p} = [u,v]_{Q,P}$$

The connection between Lagrange brackets and Poisson brackets is an inverse matrix relationship. Consider a set of $2n$ independent variables u_i ($i = 1,2,\ldots,2n$) defined on the phase space.

The relation connecting the Lagrange brackets and Poisson brackets of these variables is given by

$$\sum_{k=1}^{2n}[u_i,u_k]\{u_k,u_j\} = \delta_{ij}$$

This identity shows that the matrix composed of Lagrange brackets $[u_i,u_k]$ is the inverse matrix of the matrix composed of Poisson brackets $\{u_k,u_j\}$.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There is no Assam University question for this unit in the paper bank** — no AU Classical Mechanics paper exists in the collection, so the questions below are topic-matched from other universities' papers (non-AU) or standard syllabus-derived questions, marked as such.

Short-form questions

S1. [standard] Define the Hamiltonian of a system and write its relation to the Lagrangian.

Answer in brief: Section 2 — $H = \sum p_i \dot{q}_i - L$, with $p_i = \partial L / \partial \dot{q}_i$. For $L = T - V$ with T quadratic in velocities, $H = T + V$, the total energy.

S2. [standard] State Hamilton's equations of motion.

Answer in brief: Sections 2–3 — $\dot{q}_i = \partial H / \partial p_i$ and $\dot{p}_i = -\partial H / \partial q_i$.

S3. [standard] What is a cyclic (ignorable) coordinate? What is conserved for it?

Answer in brief: Section 4 — a coordinate that does not appear in H (or L); its conjugate momentum p_i is a constant of motion, since $\dot{p}_i = -\partial H / \partial q_i = 0$.

S4. [standard] Define a canonical transformation.

Answer in brief: Section 6 — a transformation $(q, p) \to (Q, P)$ that preserves the form of Hamilton's equations, with a new Hamiltonian $H'(Q, P, t)$.

S5. [standard] Name the four kinds of generating functions and give the relations for F_1 and F_2.

Answer in brief: Section 7 — $F_1(q, Q, t)$, $F_2(q, P, t)$, $F_3(p, Q, t)$, $F_4(p, P, t)$. For F_1: $p_k = \partial F_1 / \partial q_k$, $P_k = -\partial F_1 / \partial Q_k$. For F_2: $p_k = \partial F_2 / \partial q_k$, $Q_k = \partial F_2 / \partial P_k$.

S6. [standard] Define the Poisson bracket of two phase space functions f and g.

Answer in brief: Section 10 — $\{f, g\} = \sum_k (\partial f / \partial q_k \, \partial g / \partial p_k - \partial f / \partial p_k \, \partial g / \partial q_k)$.

S7. [standard] State the fundamental Poisson brackets.

Answer in brief: Section 10 — $\{q_j, q_k\} = 0$, $\{p_j, p_k\} = 0$, $\{q_j, p_k\} = \delta_{jk}$.

S8. [standard] Write the equation of motion of a function $F(q, p, t)$ in Poisson bracket form. When is F a constant of motion?

Answer in brief: Section 12 — $dF/dt = \{F, H\} + \partial F / \partial t$. If F has no explicit time dependence, it is a constant of motion if and only if $\{F, H\} = 0$.

S9. [standard] Define the Lagrange bracket $[u_i, u_j]$ and state its relation to the Poisson bracket.

Answer in brief: Section 14 — $[u_i, u_j] = \sum_k (\partial q_k / \partial u_i \, \partial p_k / \partial u_j - \partial p_k / \partial u_i \, \partial q_k / \partial u_j)$. The Lagrange bracket matrix is the inverse of the Poisson bracket matrix: $\sum_l [u_i, u_l] \{u_l, u_j\} = \delta_{ij}$.

S10. [standard] State the principle of least action.

Answer in brief: Section 5 — the action $I = \int (\sum p_i \dot{q}_i - H) \, dt$ is stationary, $\delta I = 0$, for the actual path. Stationary, not necessarily minimum.

Long-form questions

L1. [non-AU] Discuss the Hamiltonian dynamics and derive Hamilton's equations of motion. — St. Aloysius, M.Sc Sem-I, Classical Mechanics, Nov 2021, Part C Q19A.

What to write: Sections 1–3 in full — Legendre transformation, $p_i = \partial L/\partial \dot{q}_i$, $H = \sum p_i \dot{q}_i - L$, then the derivation from the modified Hamilton's principle (vary q and p independently, $\delta q = 0$ at the ends, integrate by parts).

L2. [non-AU] Show that the transformation $P = \frac{1}{2}(p^2 + q^2), Q = \tan^{-1}(q/p)$ is canonical. — Christ College, M.Sc Sem-I, Classical Mechanics, Nov 2025, Section C Q15.

What to write: Section 8 in full — first find the generating function $F_1(q, Q) = \frac{1}{2}q^2 \cot Q$ and check it gives $P = \frac{1}{2}(p^2 + q^2)$; then check $\{Q, P\} = 1$ by the four partial derivatives.

L3. [non-AU] Prove that Poisson brackets remain invariant under canonical transformation. State and prove the fundamental properties of Poisson brackets. — Midnapore College, M.Sc Sem-I, PHS 101.2, Q2; Christ College, Nov 2025, Section A Q2.

What to write: Sections 10–11 for the definition and the properties (antisymmetry, linearity, Leibniz rule, Jacobi identity) with the fundamental brackets $\{q_j, q_k\} = 0$, $\{p_j, p_k\} = 0$, $\{q_j, p_k\} = \delta_{jk}$; then Section 13 for the invariance, checked on the harmonic oscillator ($\cos^2 Q + \sin^2 Q = 1$).

L4. [non-AU] Show that if a function F does not depend on time and is a constant of motion, then its Poisson bracket with the Hamiltonian vanishes; state the principle of least action and derive Lagrange brackets. — St. Aloysius, Nov 2021, Part B Q14 (first half); principle of least action and Lagrange brackets are standard / syllabus-derived, no AU source.

What to write: Section 12 — $dF/dt = \{F, H\} + \partial F/\partial t$, so with $\partial F/\partial t = 0$ and $dF/dt = 0$ comes $\{F, H\} = 0$. Then Section 5 for least action, and Section 14 for the Lagrange bracket definition, its inverse relation to the Poisson bracket, and its fundamental brackets.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.